

Name: _____

Date: _____

QUADRATICS, (THE DISCRIMINANT)

LO: I can use the discriminant to determine the number and nature of roots of a quadratic equation.

What is the discriminant?

The **discriminant** of $ax^2 + bx + c = 0$ is **$b^2 - 4ac$** . It tells us how many real roots (solutions) the quadratic has.

Key results: If $b^2 - 4ac > 0$: two distinct real roots. If $= 0$: one repeated root. If < 0 : no real roots.

Quick examples

●	$x^2 + 5x + 6$: $b^2 - 4ac = 25 - 24 = 1 > 0$, two roots	positive discriminant
●	$x^2 - 6x + 9$: $b^2 - 4ac = 36 - 36 = 0$, one root	zero discriminant
●	$x^2 + x + 1$: $b^2 - 4ac = 1 - 4 = -3 < 0$, no real roots	negative discriminant
●	Discriminant = $b^2 - 4ac$	must be b^2 MINUS $4ac$

Key idea

The discriminant $b^2 - 4ac$ determines the nature of roots:
positive = 2 roots, zero = 1, negative = none.

$$\Delta = b^2 - 4ac$$

$\Delta > 0$: two roots; $\Delta = 0$: one root; $\Delta < 0$: no real roots

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - two real roots

Find the discriminant of $x^2 + 3x - 10 = 0$ and state the number of roots.

$$a = 1, b = 3, c = -10$$

$$\Delta = 3^2 - 4(1)(-10) = 9 + 40 = 49$$

$\Delta = 49 > 0$, so two distinct real roots

Two distinct real roots

Since 49 is a perfect square, the roots will be rational.

Example 2 - repeated root

Find the discriminant of $x^2 - 8x + 16 = 0$.

$$a = 1, b = -8, c = 16$$

$$\Delta = (-8)^2 - 4(1)(16) = 64 - 64 = 0$$

$\Delta = 0$, so one repeated root

One repeated root

The quadratic is a perfect square: $(x - 4)^2 = 0$.

Example 3 - no real roots

Show that $2x^2 + x + 3 = 0$ has no real roots.

$$a = 2, b = 1, c = 3$$

$$\Delta = 1^2 - 4(2)(3) = 1 - 24 = -23$$

$\Delta = -23 < 0$, so no real roots

No real roots

A negative discriminant means the parabola does not cross the x-axis.

Example 4 - finding k for equal roots

Find k if $x^2 + kx + 9 = 0$ has equal roots.

For equal roots: $\Delta = 0$

$$k^2 - 4(1)(9) = 0 \quad k^2 = 36$$

$$k = \pm 6$$

k = 6 or k = -6

Set the discriminant to zero and solve for the parameter.

YOUR TURN – PRACTICE (deliberate practice)

1. Calculating the discriminant

Find $b^2 - 4ac$ for each equation.

- a) $x^2 + 5x + 6 = 0$
- b) $x^2 - 4x + 4 = 0$
- c) $2x^2 + 3x - 5 = 0$
- d) $x^2 + 2x + 5 = 0$

2. Stating number of roots

Find the discriminant and state the number of real roots.

- a) $x^2 + 7x + 10 = 0$
- b) $x^2 - 6x + 9 = 0$
- c) $x^2 + x + 1 = 0$
- d) $3x^2 - 2x - 1 = 0$

3. Proving no real roots

Show each equation has no real roots.

- a) $x^2 + 4x + 5 = 0$
- b) $2x^2 + x + 1 = 0$
- c) $3x^2 - x + 2 = 0$
- d) $x^2 + 6x + 10 = 0$

4. Finding parameters

Find the value(s) of k for the stated condition.

- a) $x^2 + kx + 4 = 0$, equal roots
- b) $x^2 - 6x + k = 0$, equal roots
- c) $kx^2 + 2x + 1 = 0$, equal roots ($k \neq 0$)
- d) $x^2 + 4x + k = 0$, no real roots

5. Mixed discriminant problems

Use the discriminant to answer each question.

- a) Does $x^2 + 3x + 1 = 0$ have real roots?
- b) For what values of p does $x^2 + px + 25 = 0$ have one root?
- c) Show $5x^2 - 3x + 1 = 0$ has no real solutions.
- d) Find k if $2x^2 + kx + 8 = 0$ has equal roots.
- e) Does $4x^2 - 12x + 9 = 0$ have one or two roots?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $b^2 - 4ac = 0$ means one repeated root ☐

Correct answer: _____

Reason: _____

b) The discriminant is $b^2 + 4ac$ ☐

Correct answer: _____

Reason: _____

c) A negative discriminant means no real roots ☐

Correct answer: _____

Reason: _____

d) $b^2 - 4ac > 0$ means no roots ☐

Correct answer: _____

Reason: _____

e) If Δ is a perfect square, the roots are rational ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) The equation $kx^2 + (k + 3)x + 3 = 0$ has equal roots. Find the possible values of k .

Expression: _____

Simplified: _____

b) For the equation $x^2 + 2px + (3p + 4) = 0$, find the values of p for which the equation has real roots. Express your answer as an inequality.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

